Outlier Detection and Threshold Framework

KJV 1611 / KJV 1769 Semantic Drift Study

Purpose

This document defines the rules for identifying unusually high drift aligned text units using robust thresholds appropriate for skewed, heavy tailed embedding distance distributions. It drives the tail drift component of the study and produces the candidate set for human interpretability analysis.

Guiding Principle

Outlier detection uses robust non-parametric thresholds rather than Gaussian assumptions. The thresholding framework combines the median, raw median absolute deviation, and upper percentiles to classify units relative to their local distributional context.

Distance Universe

Thresholds are computed only on canonically aligned, representation eligible unit pairs. R0 and R1 thresholds are computed independently because each representation forms a separate statistical universe.

Robust Quantities

For each stratum and representation, the framework computes n, median drift, raw MAD, P95, and P99. No Gaussian scaling factor is applied to MAD because the study does not assume normality.

Threshold Definitions

- Primary Threshold: T = max(P99, median + 3 * MAD)
- Secondary Threshold: R = max(P95, median + 2.5 * MAD)

A unit is labeled STRONG_OUTLIER when its drift value meets or exceeds the primary threshold. A unit is labeled REVIEW_CANDIDATE when its drift value meets or exceeds the secondary threshold but remains below the primary threshold. All remaining units are labeled INLIER.

Stratified Threshold Hierarchy

Thresholds are computed at canon, work-group, and work levels. Classification applies the most local valid threshold according to the precedence rule: work level first, work group second, and canon level third. This avoids over or under flagging caused by a single global threshold.

Safeguards

When a stratum has fewer than 50 units, percentile estimates are treated as unstable and the threshold falls to the MAD-based component. When MAD equals zero in a non-small stratum, the primary threshold defaults to P99 and the secondary threshold defaults to P95 to avoid degenerate median based behavior.

- Primary Threshold (small sample): T = m + 3 * MAD
- Secondary Threshold (small sample): R = m + 2.5 * MAD

If MAD = 0:

- Primary Threshold (small sample): T = P99(D)
- Secondary Threshold (small sample): R = P95(D)

Interpretation Safeguard

Threshold classifications identify exceptional embedding distance behavior, not definitive historical, semantic, or theological change. Flagged passages are candidates for review and require textual interpretation.

Study Decisions

Component	Study Decision
Primary label	STRONG_OUTLIER
Secondary label	REVIEW_CANDIDATE
Default label	INLIER
Primary threshold	max(P99, median + 3*MAD)
Secondary threshold	max(P95, median + 2.5*MAD)
Precedence	Work > Work Group > Canon
Small Sample Safeguard	MAD based thresholds when n < 50
Low Dispersion Safeguard	Percentile only thresholds when MAD = 0

Expected Output Artifacts

- threshold_summary_by_scope.csv
- drift_labels_by_unit.csv
- outlier_rankings.csv